THADOMAL SHAHANI ENGINEERING COLLEGE
DEPARTMENT OF
Artificial Intelligence & Data Science

Name of Student : Siddhi Thakur

Class : SE

Division : S2

Roll Number : 122

Batch : S23

Subject : Fundamentals of Telecommunication Systems

Experiment Number : 03

Experiment Title : Simulation of signal transmission through wired media

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Experiment No. 03

Title: Simulation of Signal Transmission through Wired Media

Aim:

To study and simulate the characteristics of signal transmission through a lossy wired medium (Coaxial Cable) using LTspice and observe the effects of attenuation and propagation delay.

Software Used:

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory:

Wired Media Characteristics: In telecommunication, physical cables like Coaxial or Twisted Pair are not ideal conductors. They possess electrical properties distributed along their length: Resistance (R), Inductance (L), Capacitance (C), and Conductance (G).

The LTRA Model: The Lossy TRAnsmission line (LTRA) model in LTspice simulates real-world behavior.

Attenuation: The signal strength decreases over distance due to Resistance (R).

Distortion: The signal shape changes (rounding of edges) due to Capacitance (C) and Inductance (L).

Propagation Delay: It takes a finite amount of time for the signal to travel from the source to the destination.

Simulation Procedure:

Step 1: Component Selection and Placement

Open LTspice and create a New Schematic.

- Voltage Source: Select voltage, and place it. Right-click it, select Advanced, then PULSE. Set: Vinitial=0, Von=5, Tdelay=0, Trise=1n, Tfall=1n, Ton=0.5u, Tperiod=1u.
- Transmission Line: Press ‘P’ search for the component ‘ltline’. Select it and place it on the schematic. It will appear with the designator O1 and LTRA.
- Load Resistor: Press ‘R’, place a resistor at the output. Right click on the resistor symbol and set resistance value to 50 ohms (to match the characteristic impedance of co-axial cable).

Step 2: Defining the Cable Model

- Right-click the ltline (O1) component. In the Value field, replace LTRA by typing 'MyCoaxCable'.
- Click the '.t' button (SPICE Directive) on the top toolbar.
- In the dialog box, paste the following command exactly: .model MyCoaxCable LTRA(R=0.1 L=250n C=100p G=0 len=100)
- Place this text anywhere on your schematic area. This defines MyCoaxCable as a 100meter lossy line.

Step 3: Wiring and Grounding

- Press 'G' to place grounds. You must ground the bottom terminal of the Voltage Source, the two bottom terminals of the ltline (O1), and the Load Resistor.
- Press 'W' to wire the components. Connect all points properly.
- Use 't' to label the input wire as IN and the output wire as OUT.

Step 4: Setting up the Simulation

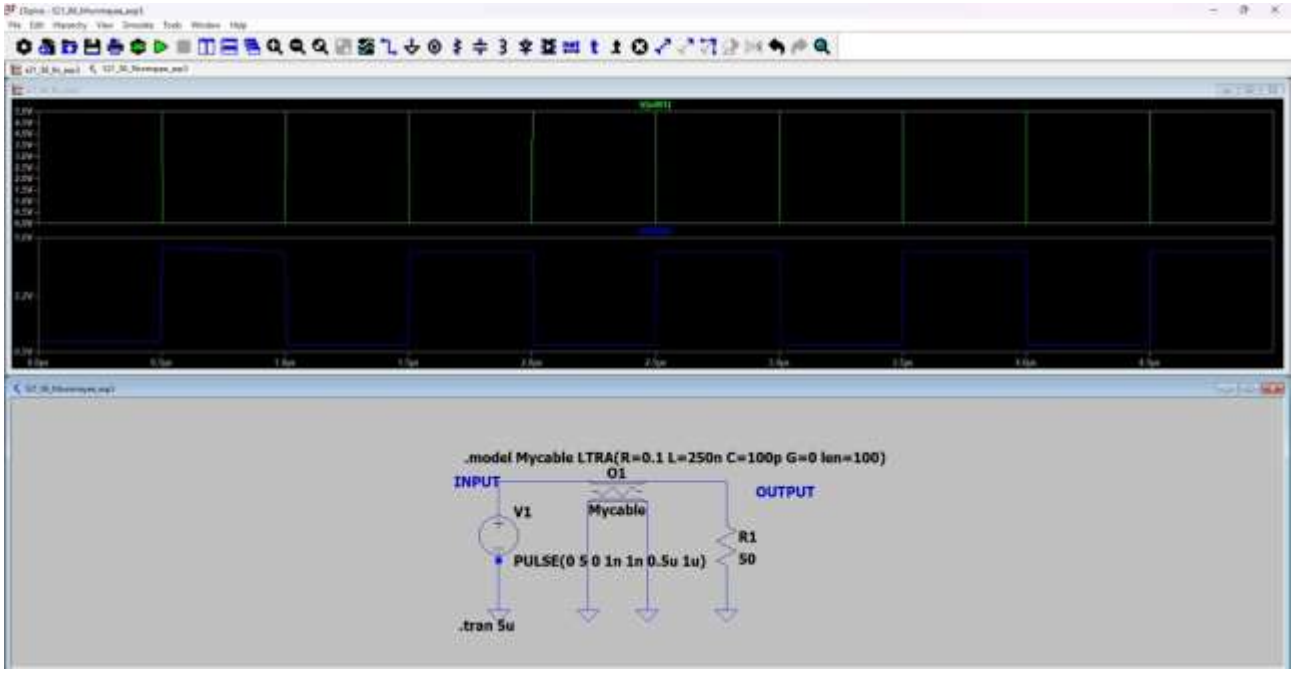
- Click configure analysis (.ac) from the toolbar or type 'A' to configure transient analysis.
- In the Transient tab, set the Stop Time to 5u.
- Place the .tran 5u command anywhere on the schematic.
- Click the Run (Running Man) icon.

Observation and Analysis :

Students must perform the simulation for the following three cases by rightclicking the .model directive and changing the len value each time.

Case 1: Cable Length = 100 meters

- Task: Run simulation with len=100.
- Screenshot: Capture the graph showing both IN and OUT waveforms.



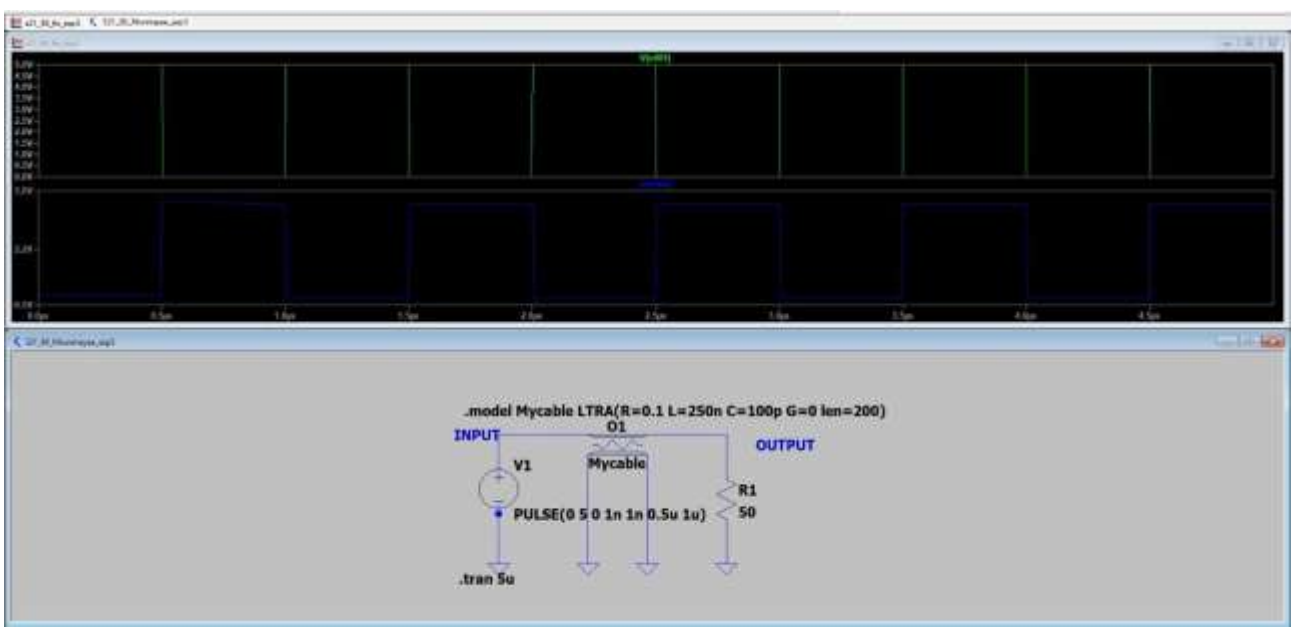
- Data: Record the peak output voltage and the time delay.

Case 2: Cable Length = 200 meters

- Task: Update directive to len=200 and re-run. Change the SPICE Directive to define cable length of 200 meters

  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 len=200)

- Screenshot: Capture the graph showing both IN and OUT waveforms.



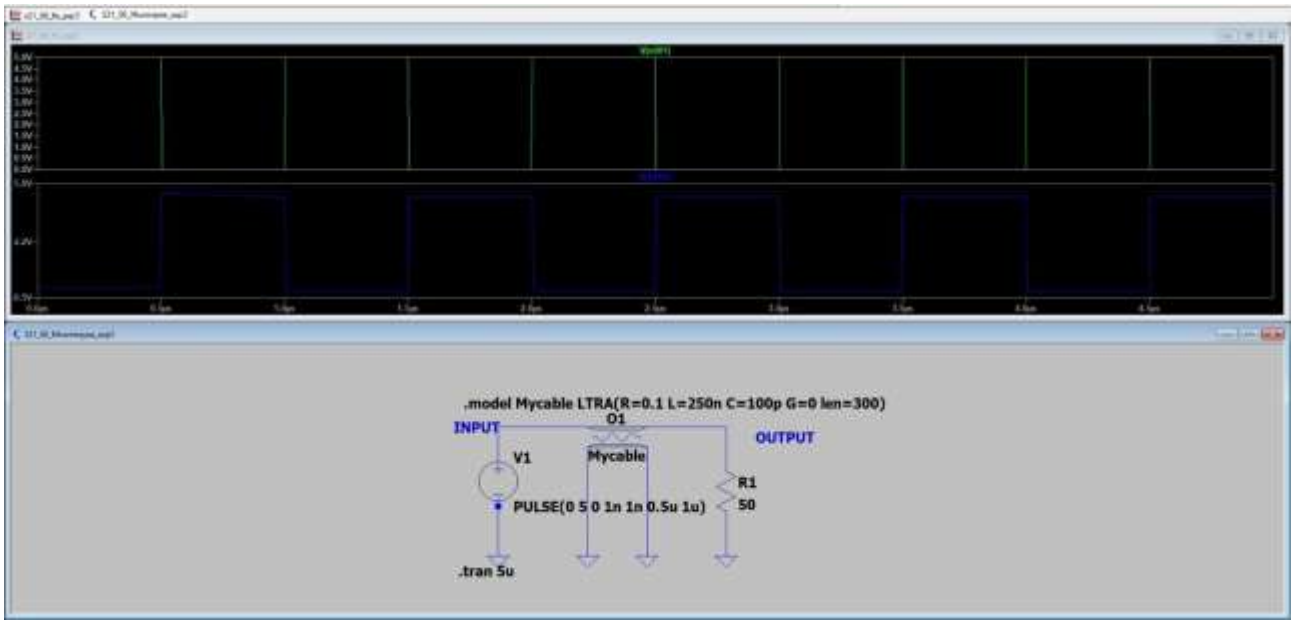
- Data: Record the peak output voltage and observe the increased delay.

Case 3: Cable Length = 300 meters

- Task: Update directive to len=300 and re-run. Change the SPICE Directive to define cable length of 300 meters

  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 len=300)

- Screenshot: Capture the graph showing both IN and OUT waveforms.

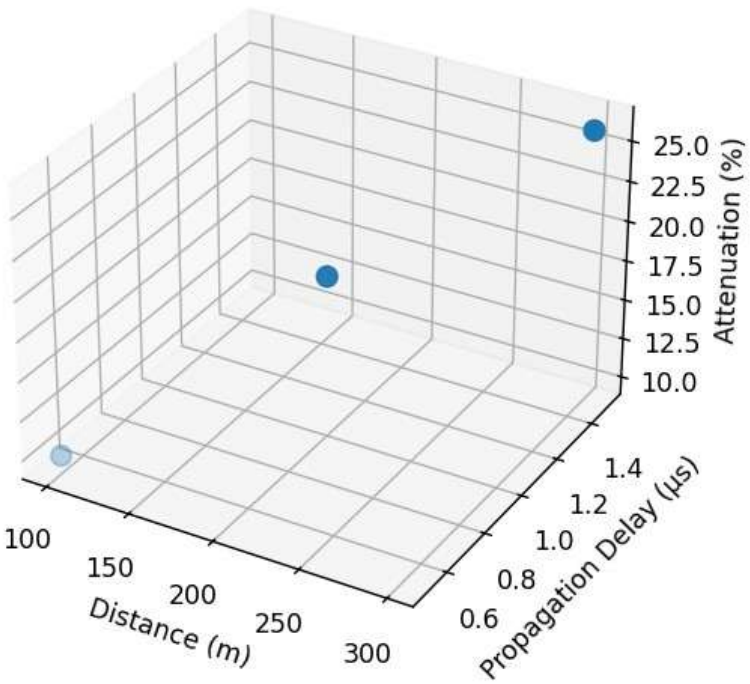


- Data: Record the peak output voltage and observe the increased delay.

  Observation Table:

Case	Cable Length	Input Voltage	Output Voltage	Propagation Delay
1	100 m	5V	4.482-4.515	0.5
2	200 m	5V	4.033-4.061	1
3	300 m	5V	3.664-3.741	1.5

Conclusion :

The simulation demonstrates that wired media is not ideal. As the cable length increases, attenuation and propagation delay become more prominent. For the cable length of 100 m, 200 m and 300 m the attenuation observed are 10.03%, 19.06%, and 25.95% respectively whereas propagation delay observed was 0.5µs, 1µs and 1.5µs respectively.

3D Plot: Distance vs Delay vs Attenuation